For Loops in PYTHON

In Python, a for loop is used to iterate over a sequence (like a list, tuple, string, or range) and execute a block of code repeatedly for each element in the sequence.

1. The Basic structure of a for Loop

A for loop allows you to repeat a block of code a fixed number of times, or once for each element in a sequence.

Syntax:

```
for item in sequence:   for each item in the sequence.
    # code to execute
```

Example:

Let's Print each name in a list of Kannad cities!

```
cities = ["Bengaluru", "mysuru", "Hubballi", "mangaluru"]
for city in cities:
    Print (city)
```

Output

```
Bengaluru
Mysuru
Hubballi
mangaluru.
```

- Here, cities is a list, and the for loop iterates over each item (city) in that list.

2. Using range() with for loops.

The range () function generates a sequence of number which you can use in a for loop when you want to repeat a block of code a specific number of times.

Syntax of range():

range (start, stop, step)

- start: The starting value (inclusive).
- stop: The ending value (exclusive).
- stop: The increment (optional, default is 1).

Example Counting from 1 to 10

```
for i in range (1.11) :
    Print (i)
```

This loop will print the numbers from 1 to 10.

Output

1
2
3
4
5
6
7
8
9
10

Example : Counting by 2s from 1 to 10

```
for i in range.(1: 11, 2) :
    Print (i)
```

This loop Prints only the odd numbers between 1 an

Output

1
3
5
7
9

3. Looping over Strings

- You can also loop over each character in a string a (?) loop.

Example: Printing each charcter in a String

```
name = "Karnataka"
for letter in name:
    Print (letter)
```

output

K
a
r
n
a
t
a
k
a

This loop goes through the string "Karnataka" one character at a time

④ Nested for Loops

You can also have nested $\boxed{\text{for}}$ loops, which means a loop inside another loop. This is useful when working with multi-level data, like lists inside lists.

Example: Multiplication Table

Let's print the multiplication table from 1 to 5 using a nested $\boxed{\text{for}}$ loop

```
for i in range (1,6):
 for j in range (1,6):
     Print (f"{i} x {j} = {i*j}")
 Print () # To Print an empty line after each table.
```

output

1×1 = 1
1×2 = 2
1×3 = 3
1×4 = 4
1×5 = 5

Here, the outer loop controls the first number (i), and the inner loop controls the second number (j). Together, they generate the multiplication table.

5. using break in a for Loop

The break statement is used to exit a loop early when a certain condition is met.

Example:- Stop the loop when you find a specific item

Let's say you are searching for a specific city in a list:

```
cities = ["Bengaluru", "mysuru", "Hubballi", "mangaluru"]
for city in cities:
    if city == "Hubballi":
        print (f"found {city}!")
        break
    print (city)
```

In this case, the loop stops when it finds "Hubballi" & prints "Found Hubballi!".

Output

Bengaluru
MySuru
Found Hubballi!